

RESEARCH ARTICLES

SOLAR CELLS

Improved solvent systems for commercially viable perovskite photovoltaic modules

Yinke Wang^{1†}, Ye Liu^{2†}, Xin Luo^{2†}, Ke Xiao^{1*}, Victor Marrugat-Arnal³, Dongdong Xu¹, Jing Lou⁴, Weiwei Zuo⁵, Navid Tavakoli⁵, Tiantian Li^{6,7}, Yuanbo Yang^{6,7}, Chenyang Duan¹, Wenna Ou¹, Yuxuan Liu¹, Jiajia Hong¹, Hongfei Sun¹, Gongtao Duan¹, Manya Li¹, Han Gao¹, Zijing Chu¹, Long Jiang^{6,7}, Michael Saliba^{5,8}, Makhud I. Saidaminov⁹, Chao Chang^{4*}, Hairen Tan^{1,2*}

Commercializing perovskite photovoltaics requires overcoming three critical barriers: the use of toxic solvents in manufacturing, variations in perovskite film quality across large areas, and limited operational reliability. Here, we address these challenges by developing a green solvent-based (γ -valerolactone/2-methyl tetrahydrofuran/dimethylsulfoxide) ink and a solvent-confinement edge-protection strategy, demonstrating large-scale manufacturing of defect-minimized perovskite films under ambient-air conditions. These approaches enabled the production of 7200-square-centimeter perovskite photovoltaic modules that achieved a total-area steady-state efficiency of 17.2% (certified by the National Renewable Energy Laboratory) and that passed all IEC 61215 reliability standards (certified by TÜV Rheinland). This work demonstrates an environmentally responsible path toward commercial manufacturing of perovskite photovoltaics.

Perovskite photovoltaics (PVs) (1–4) offer advantages in production costs (5, 6), energy efficiency (7–9), and scalability (10, 11), but commercial deployment will require the development of green, ambient-air fabrication (12–14); maximizing the total-area power output of commercial-scale modules (15); and ensuring long-term operational reliability (16). Among perovskite compositions, hybrid cation perovskites, particularly formamidinium (FA)-cesium plumbo-halides, have notable optoelectronic properties and operational stability (7, 17, 18). However, their processing typically relies on hazardous solvents such as 2-methoxyethanol (2-Me) and *N,N*-dimethylformamide (DMF) (19–21). Dimethyl sulfoxide (DMSO), although a low-hazardous alternative, is commonly paired with toxic DMF (22–24) because an excessive presence of DMSO enhances unfavorable intermediate phase dynamics during perovskite film formation. The development of green, ambient air-compatible solvent systems remains a critical unmet need for commercial-scale manufacturing of perovskite PV modules.

Scaling up perovskite modules also requires producing highly uniform films across large-area substrates. Slot-die coating offers cost-effective high-throughput film deposition at a square-meter scale (25, 26). However, inconsistent solvent coordination and extraction

kinetics across the wet film, particularly at the film edges, often lead to crystallization defects that hinder the total-area power conversion efficiency (PCE). Therefore, simultaneous optimization of the solvent chemistry and extraction kinetics is essential to overcome current performance limitations of perovskite PV modules (15).

Long-term operational reliability has been addressed in part through improved encapsulation (27–29) and interfacial engineering (30, 31). However, the intrinsic quality of perovskite films must be enhanced to improve the operational stability because bulk defects are critical determinants of device durability (32–34). Although many reports claim promising device stability on the basis of internal tests, an independent third-party verification across the full International Electrotechnical Commission (IEC) test sequences is now essential to validate the readiness of perovskite PV modules for real-world deployment (35).

We address these challenges by developing a green solvent system on the basis of γ -valerolactone (GVL) and DMSO, tuned with 2-methyl tetrahydrofuran (2-MeTHF) to tailor solvent coordination and evaporation kinetics. 2-MeTHF offers advantages in both safety and sustainability: Its methyl group reduces metabolic activation of THF into toxic by-products, and its elevated boiling point lowers inhalation risks; additionally, 2-MeTHF is derived from renewable biomass (36), unlike petroleum-based THF, and demonstrates better biodegradability and lower ecotoxicity (37). This solvent system enables the manufacturing of void-free, high-crystalline, defect-minimized perovskite films over the substrates that measure 120 cm by 60 cm. To address edge-related nonuniformities, we introduce a solvent-confinement edge-protection (SCEP) strategy to preserve solvent dynamics at the wet-film periphery. Using these approaches, we demonstrate industrially viable manufacturing of perovskite PV modules with a full area of 7200 cm². The commercial-scale perovskite PV modules achieve 17.2% steady-state efficiency, as independently certified at the National Renewable Energy Laboratory (NREL), and successfully pass all IEC 61215 reliability tests at the Technical Inspection Association TÜV Rheinland. These results establish a commercially viable and environmentally responsible pathway toward the deployment of perovskite PV modules at large scale.

Green solvents for slot-die coating

We chose FA_{0.95}CS_{0.05}PbI₃ perovskite as the light absorber because of its excellent optoelectronic properties and operational stability. The inclusion of cesium enhances the crystallinity and crystal lattice stability of the perovskite (7). To dissolve FA_{0.95}CS_{0.05}PbI₃ perovskite using green solvents, we first attempted to optimize the DMSO/acetonitrile/ethanol solvent system that we recently used in wide-bandgap perovskite (Cs_{0.35}FA_{0.65}PbI_{1.8}Br_{1.2}) fabrication (14). A DMSO volume fraction exceeding 30% was required for complete dissolution of FA_{0.95}CS_{0.05}PbI₃ (fig. S1). However, this high DMSO content not only suppressed nucleation but also strongly coordinated with the perovskite precursors, hindering its removal during crystallization. As a result, the films exhibited poor interfacial adhesion, ultimately causing delamination from the substrates (fig. S2). The high cesium and bromide content in wide-bandgap perovskites was likely responsible for promoting high-rate nucleation in our previous work (38), thereby facilitating the formation of dense and homogeneous films, which was not the case for the 1.5-eV perovskites in this work.

The 1.5-eV perovskite precursors exhibited good solubility in polar aprotic solvents with weak coordination, such as DMF and 2-Me, but

¹National Laboratory of Solid State Microstructures, College of Engineering and Applied Sciences, Frontiers Science Center for Critical Earth Material Cycling, Jiangsu Physical Science Research Center, Nanjing University, Nanjing, China. ²Research and Development Center, Renshine Solar (Suzhou) Co., Ltd., Changshu, Jiangsu, China. ³Department of Electrical and Computer Engineering, University of Victoria, Victoria, BC, Canada. ⁴Innovation Laboratory of Terahertz Biophysics, National Innovation Institute of Defense Technology, Beijing, China. ⁵Institute for Photovoltaics, University of Stuttgart, Stuttgart, Germany. ⁶State Key Laboratory of Oil and Gas Equipment, CNPC Tubular Goods Research Institute, Xi'an, China. ⁷Shaanxi Provincial Key Laboratory of Clean Energy Equipment and Materials, Xi'an, China. ⁸Helmholtz Young Investigator Group FRONTRUNNER, IMD3-Photovoltaik, Forschungszentrum Jülich, Jülich, Germany. ⁹Department of Chemistry, University of Victoria, Victoria, BC, Canada. *Corresponding author. Email: kexiao@smail.nju.edu.cn (K.X.); gwylyzssb@pku.edu.cn (C.C.); hairentan@nju.edu.cn (H.T.) †These authors contributed equally to this work.

they were unsuitable for ambient-air mass production because of their biotoxicity. By contrast, GVL has lower biotoxicity (24) and has been used for slot-die coating perovskites (39), although cesium-containing compositions have shown limited solubility. The addition of DMSO was hence required to achieve complete dissolution, which also improved the colloidal stability of the precursor ink (fig. S3).

For scalable manufacturing of perovskite solar modules at a square-meter scale, we used a combination of slot-die coating and vacuum chamber drying (VCD) (Fig. 1A) (40, 41). During the VCD, pressure and evacuation speed critically affect perovskite film formation. Low pressure accelerates solvent removal, enhancing precursor supersaturation and promoting the nucleation of fine, dense crystal nuclei. However, excessively fast evacuation may result in small, uneven grains, along with pinholes or cracks. By contrast, slower evacuation

enables gradual solvent evaporation, facilitating the growth of larger grains and reducing grain boundary defects.

Although VCD is designed to extract solvents, the residues of solvents with a high boiling point and low saturated vapor pressure (such as DMSO and GVL; fig. S4) remain in the film (34, 42), resulting in voids at the buried interface of perovskite with fluorine-doped tin oxide (FTO)/ NiO_x (Fig. 1B). To address this issue, 2-MeTHF, another green solvent with high vapor pressure and weak coordination ability to Pb^{2+} , as shown in solubility tests (fig. S5), was used (43). We found that the introduction of 2-MeTHF facilitates GVL evaporation to enhance the buried interface bonding, effectively eliminating voids at the perovskite-substrate interface (Fig. 1B). Because of the low boiling point and high saturated vapor pressure of 2-MeTHF, the entire solvent system can be extracted quickly, leading to the supersaturation of the perovskite wet film,

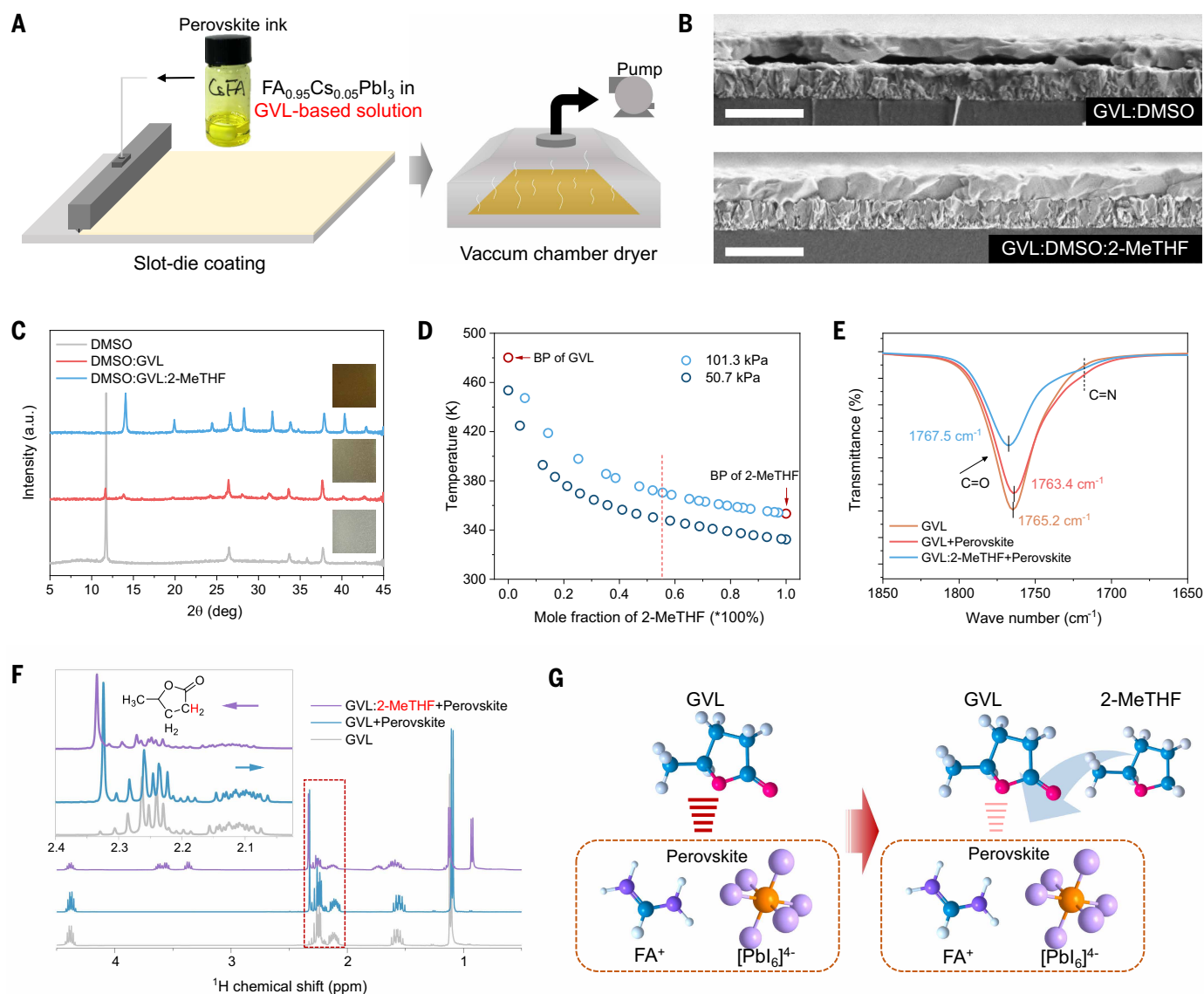


Fig. 1. Perovskite preparation with GVL/DMSO/2-MeTHF mixed solvent system. (A) Schematic demonstrating large-scale fabrication of perovskite films with slot-die coating and VCD. Digital image of a perovskite ink composed of FA_{0.95}CS_{0.05}PbI₃ dissolved in GVL/DMSO mixture. (B) Cross-sectional SEM of the perovskite films fabricated from GVL/DMSO (top panel) and GVL/DMSO/2-MeTHF mixtures (bottom panel). The scale bars represent a length of 1 μm. (C) XRD patterns of intermediate perovskite films after VCD processing (insets show digital images of the films). (D) Isobaric vapor-liquid equilibria of GVL/2-MeTHF mixture at $p = 101.3$ and 50.7 kPa (43). The red dashed line represents the optimized molar ratio of 2-MeTHF (~0.545). BP, boiling point. (E and F) FTIR spectra (E) and ¹H-NMR spectra (F) of perovskite dissolved in GVL and GVL/2-MeTHF. (G) Schematic showing coordination of GVL and 2-MeTHF to perovskite.

improved nucleation dynamics, and ultimately, reduced presence of voids at the buried interface. The increased volatility of 2-MeTHF also enabled a wider processing window under VCD. Therefore, despite varying VCD durations, the 7200-cm² perovskite films exhibited consistent quality (fig. S6), which can be translated to area tolerance, too.

With the volume ratio of 2-MeTHF increasing up to 70% in the perovskite ink, the precursor solution began to precipitate (fig. S7). The perovskite ink with a 2-MeTHF volume ratio <70% demonstrated excellent stability after 10-day aging under ambient-air conditions, rendering it suitable for high-volume disposable preparation in manufacturing. This excellent ink stability is largely attributed to GVL, which can strongly interact with FA⁺ cations (39). Upon the addition of 2-MeTHF into perovskite ink, colloidal particles increased in size as shown by dynamic light scattering, and PbI₂-DMSO and PbI₂-GVL complexes weakened as shown by ultraviolet (UV)-visible spectra (fig. S8) (44).

Scanning electron microscopy (SEM) analysis of perovskite films revealed a progressive enlargement of the average grain size with increased 2-MeTHF content (fig. S9). We reason that introducing the volatile and noncoordinating 2-MeTHF rapidly reduces the precursor solubility, thereby promoting nucleation of the desired α phase while simultaneously suppressing δ phase. Meanwhile, residual DMSO facilitates the growth of the α -phase nuclei (fig. S10). Through characterization of film crystallinity and optical properties, we identified that the perovskite film fabricated with a 50% volume ratio of 2-MeTHF demonstrated optimal properties, exhibiting both the highest crystallinity and the strongest light absorption (fig. S11). Although the perovskite ink with 70% 2-MeTHF produced large grains, it also exhibited a large fraction of nonperovskite phase, likely because this ink was at a critical threshold at which the ink becomes turbid.

To elucidate the crystallization mechanism mediated by 2-MeTHF, we analyzed x-ray diffraction (XRD) patterns of intermediate perovskite films fabricated with different solvents immediately after VCD processing (Fig. 1C). The intermediate film prepared with pure DMSO exhibited a fully coordinated complex, whereas the DMSO/GVL system produced a mixture of the complex and α -phase perovskite micronuclei. The higher volatility facilitated by the introduction of 2-MeTHF into the DMSO/GVL system induces supersaturation rapidly and appears to increase the nucleation barrier for δ phase (fig. S10). As a result, α -phase nuclei can grow into phase-pure high-quality crystals during subsequent annealing.

To reveal how the incorporation of 2-MeTHF can accelerate perovskite nucleation, we examined the isobaric vapor-liquid equilibria of GVL/2-MeTHF and identified that the introduction of 2-MeTHF substantially depressed the boiling point of GVL, thereby enhancing the volatility of the solvent system (Fig. 1D) (45). Interestingly, this temperature is slightly higher than that of pure 2-MeTHF, suggesting that 2-MeTHF facilitated GVL volatilization. We then used Fourier transform infrared (FTIR) spectroscopy to investigate solvent-perovskite interactions (Fig. 1E). Compared with pure GVL, the perovskite-GVL solution exhibited weakened C=O bond resonance along with a blue-shifted absorption peak, indicating coordination between the carbonyl group of GVL and the perovskite building blocks. However, when 2-MeTHF was introduced, the C=O resonance has redshift, suggesting weakened interaction with perovskite.

Proton nuclear magnetic resonance (¹H-NMR) spectra (Fig. 1F and fig. S12) of perovskite inks revealed upfield shifts for both the CH proton of GVL and the NH proton of FA⁺ when dissolved together, compared with pure GVL or FAI, indicating interaction between the FA⁺ cation and GVL. A similar behavior was found when incorporating GVL to DMSO-only ink, indicating even stronger solvent-precursor interactions. However, upon incorporating 2-MeTHF, these proton signals shifted downfield, indicating a weakening of FA⁺-GVL interactions. In Fig. 1G, we propose 2-MeTHF incorporation mechanism derived from the discussed experimental results: 2-MeTHF reduces perovskite-GVL coordination, thereby promoting GVL extraction during VCD and annealing,

which then efficiently improve interfacial contact and crystallization of the perovskite films.

Solvent-confinement edge protection

We observed that when the perovskite ink was coated on the substrate before VCD-assisted solvent extraction, a spontaneous volatilization of solvent at the gas-solid-liquid boundary triggered crystallization at the edges of the film (fig. S13). This phenomenon is likely caused by the rapid volatilization induced by 2-MeTHF incorporation, which narrows the processing window, as seen by shrinkage of absorption profile of intermediate phase (fig. S14); this makes it challenging to achieve a consistent film quality (46, 47). Such morphological defects pose a challenge in the fabrication of large-area perovskite films, as the poor crystallinity at the periphery adversely affects the total-area performance of commercial PV modules. Therefore, we pursued the SCEP strategy by adding trimethyl-tetradecylammonium chloride (TAC) into the perovskite ink. TAC, a surfactant molecule containing both hydrophilic and hydrophobic functional groups, can self-assemble at the solution's surface, thereby modulating the ink's surface tension and the crystallization kinetics (48).

To investigate the surface tension and the evolution of solvent volatilization, a perovskite ink droplet was deposited on an FTO/NiO_x substrate under ambient conditions (Fig. 2A). The initial contact angle of the perovskite ink with TAC was lower than that of the control ink, indicating a reduction of surface tension by TAC. During the initial stage of droplet deposition, interfacial wetting at the solid-liquid boundary dominated over solvent evaporation. As the solid-liquid interface progressively stabilized, the predominant mechanism shifted to solvent evaporation at the liquid-vapor interface. Because of the rapid evaporation at the edge of the control droplet, solution concentration gradually increased at the periphery, leading to an increased contact angle. By contrast, the droplet with TAC exhibited suppressed edge evaporation, demonstrating uniform volatilization across the entire gas-liquid interface and a progressively decreasing contact angle (fig. S15). By orienting hydrophobic groups toward the air, the TAC molecules balanced solution flow from the droplet's surface to the droplet's bulk through surfactant-induced Marangoni effect, thus leading to uniform areas at the edges of large-area films (48, 49). This phenomenon also delayed solvent volatilization at the gas-liquid-solid boundaries (Fig. 2B).

X-ray photoelectron spectroscopy (XPS) of SCEP perovskite films exhibited a C 1s peak at 286.39 eV (fig. S16), attributed to long carbon chains, along with a concomitant N 1s peak at 402.51 eV, corresponding to the amine group from TA⁺. No detectable Cl XPS signal was observed on the surface. Time-of-flight secondary ion mass spectrometry revealed a significant reduction in TA⁺ distribution from the top surface to the bottom surface of perovskite films. The graded distribution of TA⁺ residues from the top surface to the bottom interface of perovskite indicates that TA⁺ is predominantly localized at the perovskite surface, where it passivates defects and thus enhances film quality.

To verify the influence of the SCEP strategy on charge-carrier mobility and diffusion length in perovskite films, we performed femtosecond-resolved optical-pump terahertz-probe spectroscopy (50). Terahertz photons with energies below the semiconductor bandgap allow the exclusive investigation of free-carrier dynamics by circumventing interband excitation. Concomitantly, the picosecond-period terahertz field drives carriers efficiently when its oscillation frequency resonates with the characteristic momentum relaxation rate. The resultant carrier acceleration is limited by the rate of energy dissipation via inelastic scattering. Figure 2, C and D, illustrates the transient terahertz transmission dynamics for control and SCEP thin films (~600-nm thickness) under varying pump fluences. An 800-nm (1.55-eV) laser with a ~5-mm spot diameter served as the optical pump, and sub-picosecond terahertz pulses probed photoinduced conductivity changes. By using a reduced optical-pump energy in conjunction with terahertz probe

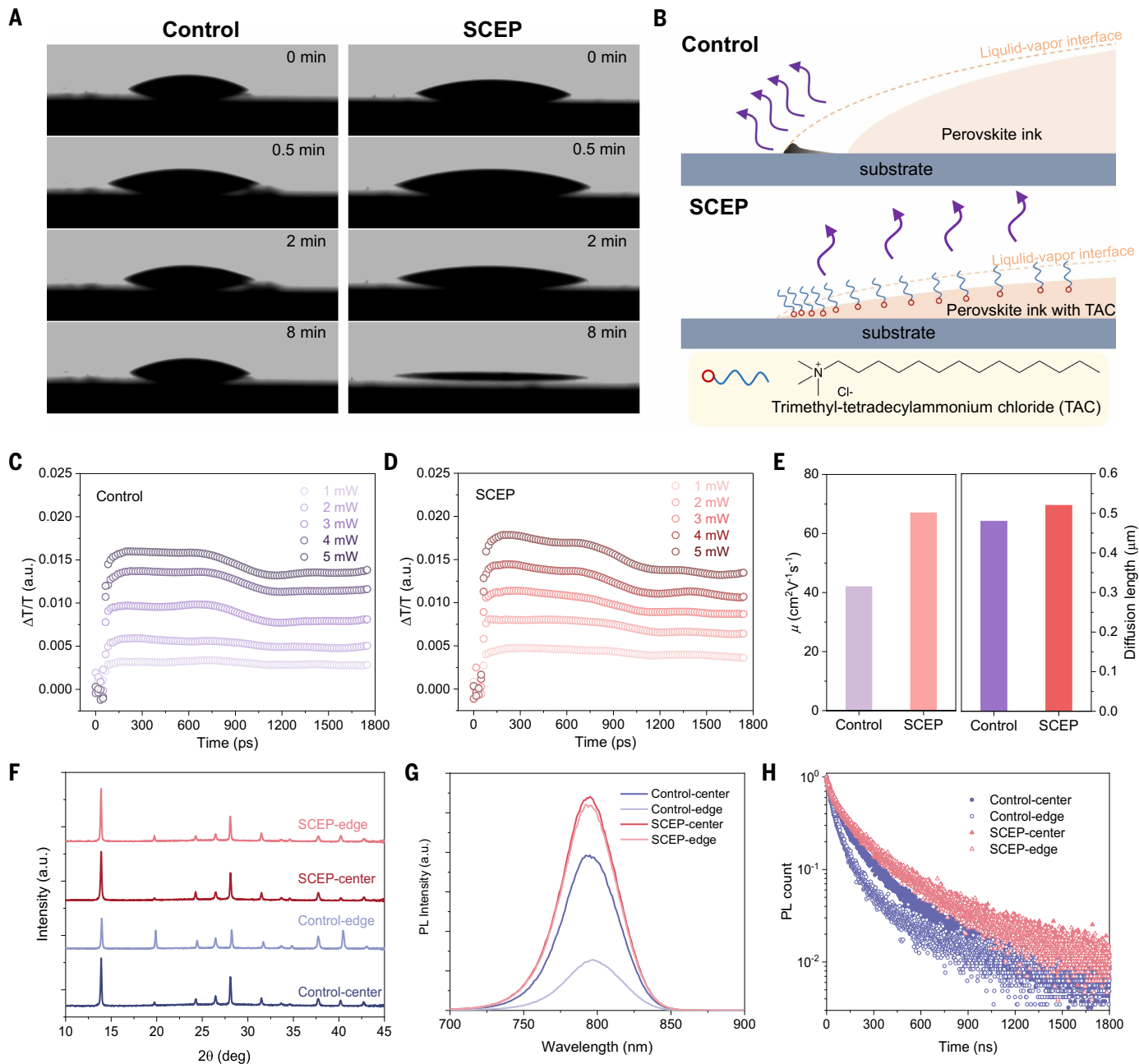


Fig. 2. Characterization of perovskite films. (A) Contact-angle images of the control and SCEP perovskite ink droplets. (B) Schematic illustrating solvent evaporation of the perovskite wet film. (C and D) Free-carrier dynamics of the control (C) and SCEP (D) perovskite films based on terahertz measurement. (E) Mobilities (left) and diffusion lengths (right) of the control and SCEP perovskite films based on terahertz analysis. (F to H) XRD patterns (F), steady-state PL spectra (G), and time-resolved PL traces (H) of control and SCEP perovskite films from center and edge regions.

radiation, we enhanced the accuracy in determining carrier mobility and diffusion length parameters. This approach minimizes thermal perturbations and associated hot-carrier effects during the measurements. The recombination rate constants and charge-carrier mobilities for control and SCEP samples were calculated to be 42 and 67 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$, respectively (supplementary text 1, fig. S17, and table S1). Terahertz time-resolved spectroscopy revealed longer carrier lifetimes in the SCEP film, leading to an increased diffusion length of 0.54 μm extracted from the terahertz measurements, compared with 0.48 μm for the control film (Fig. 2E). These findings suggest that the SCEP perovskite film demonstrated improved crystallization quality and reduced nonradiative recombination.

The perovskite ink prepared by SCEP strategy exhibited two benefits: (i) substantial extension of the processing window during ink deposition before VCD (fig. S18) and (ii) significant improvement in film uniformity through effective suppression of defective edge formation (fig. S19). SEM images showed poorly defined surface morphology in the central region of a 120-cm-by-60-cm control perovskite film and small grains accompanied by voids on the edges (fig. S20). Atomic force microscopy similarly showed an increased surface roughness at the periphery of the control sample (fig. S21). The SCEP strategy instead resulted in a similar morphology within all regions of the target film, as well as an overall increase of grain size. The perovskite film fabricated with the SCEP strategy demonstrated notable thickness uniformity,

exhibiting an average thickness of 393 nm with a thickness variation of only 6.5% across the entire 30-cm-by-40-cm substrate area. By contrast, the control film showed substantial thickness reduction (average thickness of 338 nm), particularly at the substrate edges (fig. S22).

We then compared the crystallinity uniformity and photophysical properties of the control and SCEP perovskite films. The SCEP perovskite film demonstrated consistent crystallinity and a predominant (100) crystallographic orientation throughout both the central and edge regions. By contrast, the control film exhibited reduced crystallinity and disordered orientation at the edge region (Fig. 2F). Although we did not find that TAC promotes the formation of low-dimensional perovskites, it still enhanced the crystallographic orientation of perovskite films (fig. S23). To investigate the photophysical properties of the control and SCEP perovskite films, we compared the steady-state photoluminescence (PL) and time-resolved PL lifetimes at the central and edge regions. The steady-state PL emission of the SCEP perovskite film measured from both central and edge regions were identically strong and higher than those of

the control film (Fig. 2G). The PL decay lifetimes (τ) probed from both regions of the SCEP perovskite film were similar ($\tau = 277$ ns from center and $\tau = 247$ ns from edge) and longer than those of the control film ($\tau = 186$ ns from center and $\tau = 130$ ns from edge) (Fig. 2H and table S2).

Performance of PV modules

Perovskite PV modules were fabricated in a FTO/ NiO_x /perovskite/ C_{60} / SnO_2 /Cu architecture. For this, perovskite films were first deposited on PI-patterned FTO/ NiO_x substrates, followed by the deposition of charge-selective layers, including fullerene (C_{60}) and tin oxide (SnO_2). Subsequent processing steps involved laser scribing for cell interconnection, electrode deposition, and encapsulation. The perovskite layer was fabricated under ambient conditions, whereas other functional layers were deposited via vacuum processes (fig. S24).

To reconfirm the optimum content of 2Me-THF in perovskite inks, we made 1017.5-cm^2 control modules and measured their performance (fig. S25). In line with studies described above, the perovskite ink with

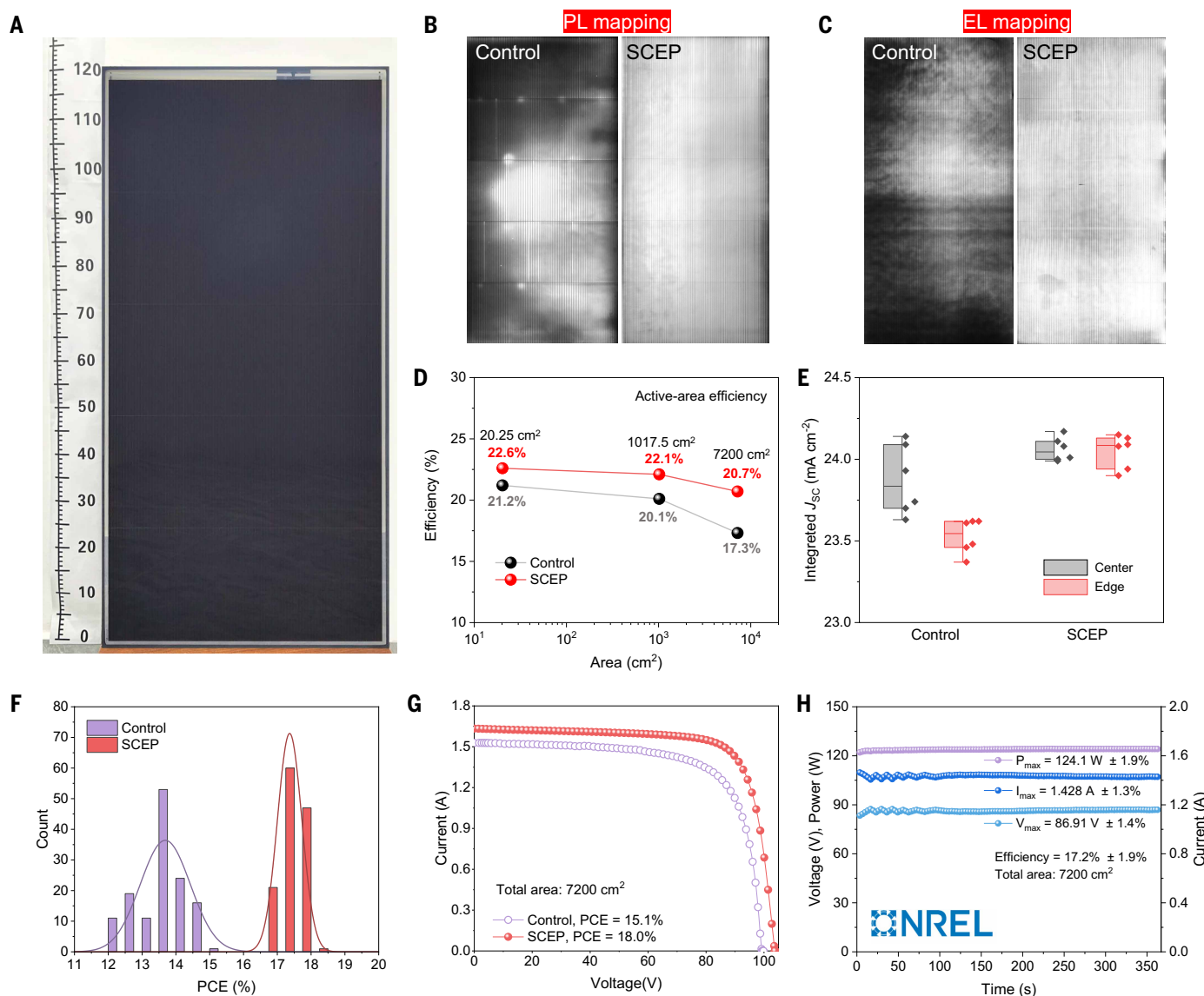


Fig. 3. Performance of commercial-scale perovskite PV modules (120 by 60 cm²). (A) Image of a module. (B and C) PL (B) and EL (C) mapping of control and SCEP modules. (D) Comparison of active-area efficiencies between the champion 20.25-, 1017.5-, and 7200-cm² modules. (E) Comparison of the integrated J_{sc} values from center and edge regions in control and SCEP modules. (F) Statistical histogram of total-area module PCEs for control and SCEP modules. (G) J-V curves of the best-performing control and SCEP modules with a total area of 7200 cm². (H) NREL-certified stabilized P_{max} , I_{max} , and V_{max} plots within 360 s for a module.

50% 2Me-THF volume ratio showed the highest PCE. We then compared TAC with other surfactants, such as sodium dodecyl sulfate, Tween 60, and Triton X-100, all of which achieved the desired uniformity (figs. S22 and S26) and similar crystallinity (figs. S23A and S26C), but TAC remained the champion surfactant (fig. S26D). We hypothesize that this is due to the cationic nature of TAC, which allows it to perform passivation functions while regulating the wet-film flows (51–53). Through systematic study of TAC loading in the perovskite ink, we found a TAC concentration of 0.2 g liter^{−1} to show the highest PV performance (fig. S27).

Figure 3A shows an encapsulated 120-cm-by-60-cm commercial-scale perovskite PV module. To evaluate the effect of SCEP in these modules, we conducted PL and electroluminescence (EL) mapping (Fig. 3, B and C). The results demonstrate that the SCEP strategy achieved more uniform and enhanced radiative emission compared with the control modules.

The SCEP perovskite modules with aperture areas of 20.25 and 1017.5 cm² achieved champion PCEs of 21.9 and 21.4%, respectively (fig. S28), whereas the control modules achieved PCEs of 20.6 and 19.5%. The best-performing commercial-scale module achieved a PCE of 18.0% with a full area of 7200 cm² as measured in the laboratory (Table 1). Given the geometric fill factors (FFs) of 97% for both 20.25- and 1017.5-cm² modules and 87.1% for 7200-cm² modules, the corresponding active-area PCE values were calculated to be 22.6, 22.1, and 20.7%, respectively (Fig. 3D). This minimal performance gap, despite substantially upscaling the module areas, demonstrates the scalability of the technology from laboratory- to commercial-scale PV modules.

The integrated short-circuit current density (*J*_{sc}) values of the control and SCEP modules were similar (24.1 versus 24.2 mA cm^{−2}), as derived from external quantum efficiency spectral analysis (fig. S29). However, when comparing the integrated *J*_{sc} between the central and edge areas (fig. S30), the SCEP modules showed improved spatial uniformity over the control modules (Fig. 3E), directly enhancing the performance and reproducibility of large-scale PV modules (fig. S31).

We conducted a statistical analysis of 127 PV modules using the SCEP strategy manufactured within a single production day, revealing a normal distribution of output power with an average value of 125 W (fig. S32). Statistical comparison between the control and SCEP modules revealed that the SCEP devices exhibited significant performance enhancement as well as improved reproducibility (Fig. 3F). The best-performing PV characterization results, as presented in Fig. 3G, revealed that the SCEP-optimized module achieved a full-area PCE of 18.0% under reverse scan conditions, with the following corresponding parameters: open-circuit voltage (*V*_{oc}) of 104 V, short-circuit current (*I*_{sc}) of 1.63 A, and FF of 76.5%. By contrast, the control module exhibited a lower PCE of 15.1%. After testing in the lab, we sent the 18.0% efficiency SCEP module to NREL, where it achieved a certified stabilized efficiency of 17.2% with a full device area of 7200 cm² (Fig. 3H), designating it a total-area certified commercial perovskite PV module (fig. S33) (54).

Table 1. PV performance of control and SCEP perovskite modules with aperture area of 1017.5 cm² or total area of 7200 cm². Module number of control and SCEP for the error bars are 135 and 129, respectively. Means are shown with standard deviations.

		<i>V</i> _{oc} (V)	<i>I</i> _{sc} (A)	FF (%)	PCE (%)
1017.5 cm ² (aperture area)	Control	47.8 ± 0.4	0.547 ± 0.003	71.4 ± 2.3	18.4 ± 0.75
	SCEP	49.2 ± 0.1	0.562 ± 0.002	77.0 ± 0.8	20.9 ± 0.29
7200 cm ² (total area)	Control	94.1 ± 4.0	1.500 ± 0.025	69.9 ± 2.7	13.7 ± 0.74
	SCEP	104.0 ± 1.1	1.620 ± 0.013	74.2 ± 1.2	17.4 ± 0.36

Reliability of the perovskite PV modules

IEC 61215 is an international standard in PV industry, primarily designed to assess the reliability and safety of commercial PV modules. Leveraging optimized edge properties and enhanced crystallization quality, SCEP modules exhibited significant improvements across all reliability metrics evaluated under IEC standards. Thus, we have completed all required tests by these standards, as certified by the Technical Inspection Association TÜV Rheinland (fig. S34).

Although long-term maximum power point tracking (MPPT) stability is not included in the IEC 61215 test sequences, it serves as a critical indicator of module reliability under real-world operating conditions. As demonstrated in Fig. 4A, SCEP modules exhibited enhanced operational stability, retaining 97.2% of their initial PCE after 1050 hours of continuous MPPT operation under 1 sun illumination (AM1.5G) at the module temperature of ~60°C (fig. S35). By contrast, control modules experienced substantial degradation, losing nearly 20% relative to their initial PCE under identical testing conditions. We also performed additional MPPT testing under a controlled steady temperature (65°C) (fig. S36): The SCEP module again showed a superior stability when compared with control after 200 hours.

Then, we assessed the devices against all tests specified by the IEC 61215 standards, including the damp heat test, thermal cycling, and ultraviolet (UV) preconditioning test, among others (fig. S37 and supplementary text 2). The SCEP module maintained its initial performance after an outdoor exposure test with a cumulative solar irradiation of 70 kilowatt-hours (kWh) m^{−2} (fig. S38). Under UV preconditioning conditions, the SCEP module maintained 96.1% of its initial performance after exposure to 36 kWh m^{−2} of total solar irradiation (Fig. 4B), exceeding the IEC 61215 standard requirement by more than twofold. By contrast, the control module maintained only 72.1% of their initial PCE under identical test conditions.

Figure 4C presents the results of the thermal cycling test conducted between −40° and +85°C, with current flow. The SCEP modules exhibited no degradation in the thermal cycling test without current and retained 97.5% of its initial performance after 200 cycles with current (fig. S39). With current, the control modules exhibited substantial degradation after 50 cycles, with their PCE dropping to only 69% of the initial value. Figure 4D presents damp heat test results (85°C and 85% relative humidity), in which the SCEP module maintained 100% of its initial PCE after 1000 hours. Control modules, however, degraded to 73.4% PCE after 1000 hours and further declined to 60.9% after 2000-hour aging (fig. S40).

Post-aging characterization revealed edge-initiated degradation in control modules (fig. S41), indicating that the initially visible poor crystal formation of the perovskite film at module edges accelerates performance deterioration. We further characterized the modules through EL mapping before and after damp heat, MPPT, thermal cycling, UV tests, and light-induced degradation aging tests (fig. S42). The control modules exhibited significant contrast variations in their EL images, with these pronounced changes directly correlating to their observed performance degradation. We hence hypothesize that the main degradation mechanism in control PV modules is poor crystallization and uniformity, especially on the edges.

We also evaluated the stability of perovskite/C₆₀ interfaces by exposing these stacks to ambient air for 6 hours (fig. S43): The reduced absorption and the appearance of δ-phase peak in the control film indicate that it underwent a noticeable degradation; by contrast, the film including TAC remained stable under the same exposure.

The temperature coefficient of efficiency for the perovskite PV modules was measured at −0.11% per °C, versus an approximate range of −0.29 to −0.39% per °C for crystalline silicon PV modules (Fig. 4E) (55). Figure 4F illustrates the results of the hot-spot endurance test. The performance of the module decreased to 87.6% of its initial efficiency after a 3-hour shading condition but subsequently recovered to its original efficiency, demonstrating its robust self-healing capability, which enables the

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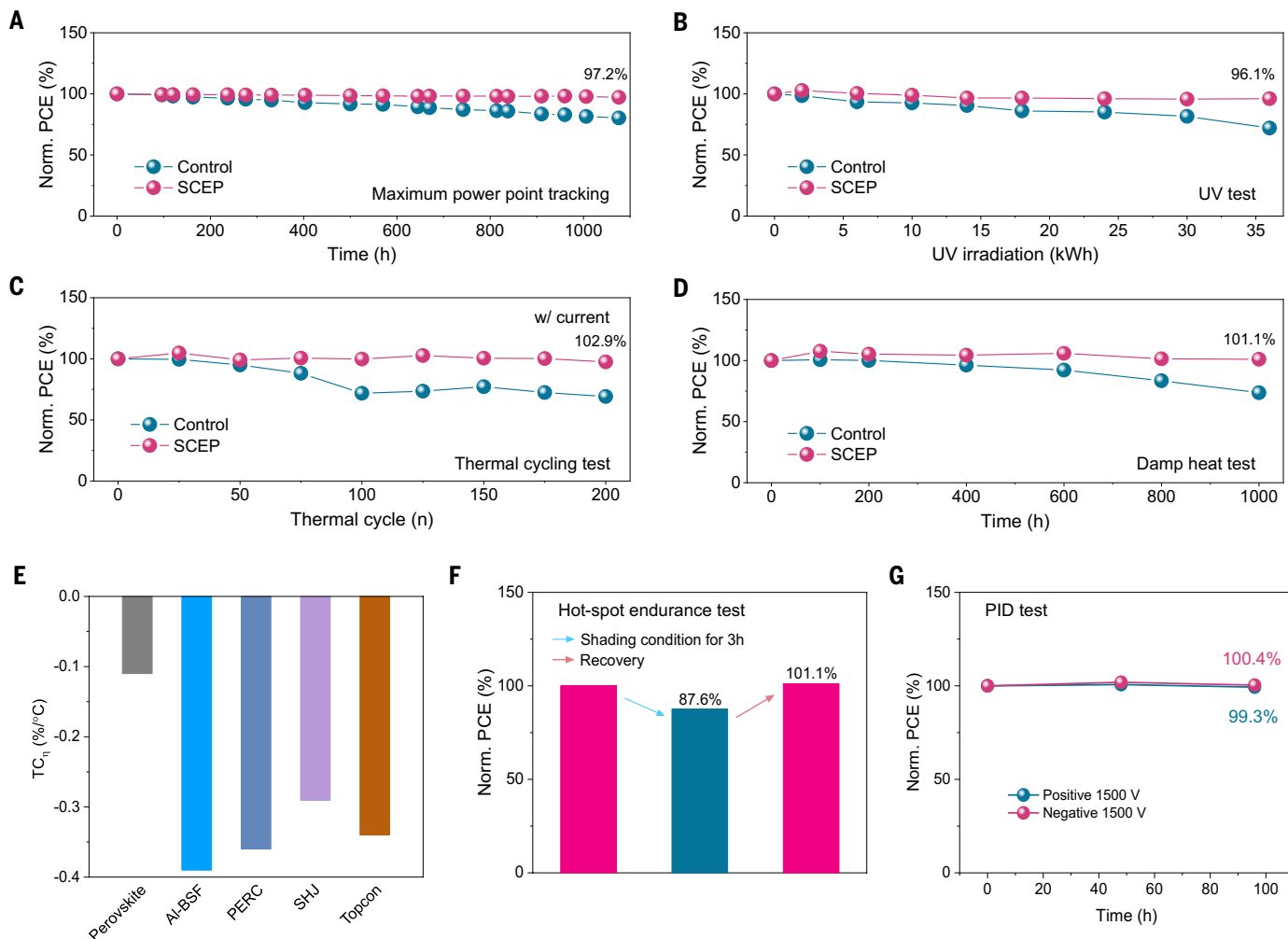


Fig. 4. Reliability tests of commercial perovskite PV modules following IEC 61215:2021 standards. (A) Continuous MPPT of encapsulated modules over 1000 hours under full simulated AM1.5G illumination in ambient air with a humidity of 30 to 50%. The modules reached 60° to 65°C during the MPPT. (B) UV preconditioning test with a cumulative UV irradiation of 36 kWh m⁻². (C) Thermal cycling test from -40° to 85°C with or without current. (D) Damp heat test at 85°C and 85% RH conditions. RH, relative humidity. (E) TC comparison between perovskite modules and various types of crystalline silicon modules (56,57). Al-BSF, aluminum back surface field; PERC, passivated emitter and rear cell; SHJ, silicon heterojunction; TC, temperature coefficient. (F and G) Hot-spot endurance test (F) and PID test (G) of commercial-scale perovskite modules.

modules to effectively handle real-world challenges such as temporary shadowing from surrounding objects.

Additionally, potential-induced degradation (PID), a newly added requirement in the IEC 61215:2021 standard, was evaluated. We applied a high voltage of 1500 V, both negative and positive, to the module for >90 hours (Fig. 4G). The results showed no significant degradation, confirming the module's ability to maintain safe, stable, and efficient operation in PV power plants under high-voltage stress conditions.

We continuously monitored the daily power generation performance of the perovskite PV power station from October 2024 to August 2025 (fig. S44): The system demonstrated notably stable power output. We compared one perovskite panel within the PV station with a tunnel oxide passivated contact (TOPCon) silicon PV panel from November 2024 to August 2025. Both panels maintained consistent power generation, but when temperatures exceeded 30°C in the summer, the perovskite panel not only sustained its stable performance but showed higher power-generation output than the TOPCon panel. This comparison with a well-established technology indicates a commercial viability of perovskite solar modules.

We developed eco-friendly perovskite inks by engineering a green solvent system comprising 2-MeTHF, GVL, and DMSO, which precisely

modulates solvent coordination and evaporation kinetics. To enable large-scale fabrication, we introduced the surfactant TAC to homogenize solvent evaporation and enhance perovskite crystallization uniformity, achieving a certified 17.2% efficiency on a commercial-scale module (7200 cm²) and meeting all IEC61215 reliability standards (independently verified by TÜV Rheinland). By replacing toxic solvents (DMF/NMP) with the sustainable formulation, we simultaneously addressed environmental and scalability challenges while maintaining high performance. Furthermore, the SCEP strategy synergizes with optimized solvent volatility to deliver film uniformity at meter scale, overcoming a critical limitation of conventional approaches.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S44; Tables S1 and S2; References (58–63)

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